for a total of five dimensions in all. It shares three space dimensions with our brane: east-west, north-south, and up-down. It shares a fourth, time dimension, with our brane. And it has a fifth space dimension, *out-back*, which extends perpendicular to our brane, both above the brane and below, as depicted in Figure 21.3.

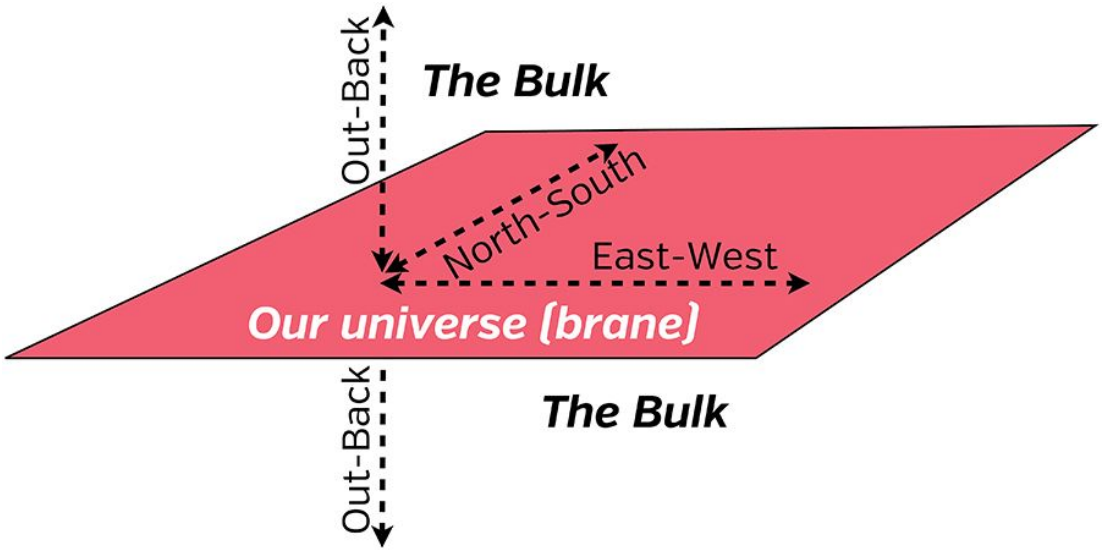


Fig. 21.3. Our universe as a brane with four spacetime dimensions, residing in a five-dimensional bulk. I have suppressed two dimensions from the diagram: time, and our universe's up-down dimension.

The out-back dimension plays a major role in *Interstellar*, though the Professor and others don't use the phrase "out-back," but instead just refer to "the fifth dimension." Out-back is central to the next two chapters, and to Chapters 25, 29, and 30.

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34 But the relativistic laws *do* offer the possibility of backward time travel by a circuitous route: going outward in space and returning before we left. To this I return in Chapter 30.

35 See Chapter 3 for a brief description of this quest.